Nano Long Range FPV Drone — Parts Breakdown

Jordan & Friend · May 2026 · Based on MakerWorld design by piodeer

Component		Weight	Dimensions	Price (est.)	Material	Notes
POWER SYSTEM						
Sony VTC6 18650 battery	✓	46.6 g	Ø18.35 × 65 mm cylindrical cell	€4–6	Steel can NMC Li-ion Nickel contacts	~47% of total AUW. CG defined by position.
18650 battery holder	est.	~5 g	~70 × 22 × 20 mm single-cell tray	€1–2	ABS plastic Ni-plated spring contacts	3D-printed PLA replacement saves ~2 g
ELECTRONICS						
1S AIO FC w/ ELRS	✓	~4–5 g	~29 × 29 mm PCB 25.5×25.5 mm holes	€12–20	FR4 fibreglass PCB STM32 MCU SMD components	FC + 4-in-1 ESC + ELRS RX on one board
VTX (nano 5.8 GHz)	est.	~2–3 g	~14–19 mm sq. varies by model	€5–12	FR4 PCB RF shielding (tin) SMD RF parts	Some AIO FCs include integrated VTX
FPV camera (nano)	est.	~3 g	~14 × 14 mm body nano form factor	€8–15	Plastic housing CMOS sensor Glass lens, PCB	FOV ~120–160 deg. nano size
LC filter	est.	~1 g	~10 × 8 × 5 mm tiny inline board	€1–3	FR4 PCB Ferrite inductor Ceramic caps	Cleans power noise to VTX / camera
5V step-up converter	est.	~2 g	~15 × 11 mm small boost module	€1–3	FR4 PCB Inductor, MOSFET SMD passives	Boosts 3.7V (18650) to 5V for VTX/cam
MOTORS & PROPS						
Motors ×4 (1202.5 11500KV)	✓	4.4 g each 17.6 g total	Ø15 × 15 mm each Stator Ø12 × 2.5 mm	€12–18 for set of 4	Aluminium bell Copper windings Steel shaft 1.5 mm N52 magnets	9N12P; 50 mm lead wires
Props ×4 (3" bi-blade)	est.	~0.5–1 g ea ~2–4 g total	~76 mm tip-to-tip varies by brand	€2–5 for set of 4	Polycarbonate (PC) or polypropylene (PP)	PC = stiffer; PP = crash-resistant
3D PRINTED FRAME						
Frame + battery tray	est.	~7–10 g	~120–130 mm WB arm-to-arm diagonal	€1–2 (filament cost)	PLA / PLA+ density ~1.24 g/cm3	Slice in Bambu Studio 20% infill / 2 walls for exact weight
HARDWARE						
M2 × 8 mm screws ×8	est.	~0.5 g	M2, 8 mm length	~€1	Stainless steel or black-anodised steel	Frame assembly
M2 × 12 mm screws ×4	est.	~0.3 g	M2, 12 mm length	~€1	Stainless steel	Motor mounting
M2 nuts + wiring	est.	~2–4 g	M2 hex nuts 28 AWG wire	€1–2	Steel nuts Silicone-insulated copper wire	Silicone wire stays flexible in cold
Total (on-drone parts only)					**~93–104 g**	**€48–87**

Confirmed (✓) = sourced from datasheet or manufacturer spec. **Estimated (est.)** = typical for component class — verify against exact parts once ordered. Prices are AliExpress baseline; Western FPV shops typically 30–50% higher. Controller (Jumper T-Lite ~€60–80) and goggles (Eachine EV100 ~€60–90) not included — ground equipment only. Frame weight: slice STLs in Bambu Studio at 20% infill / 2 walls for exact PLA weight (density 1.24 g/cm3).